Lab Sheet-9

Ashish singh rawat

Cu24250221

```
import os
import cv2
import requests
import numpy as np
import matplotlib.pyplot as plt
from ultralytics import YOLO
# Output directory path set karein
output_dir = 'experiment9'
os.makedirs(output_dir, exist_ok=True)
# Step 1: Pre-trained YOLOv8 Nano model load karein
model = YOLO('yolov8n.pt')
# Step 2: Sample test image download karke experiment9 folder me save karein
image_url = 'https://ultralytics.com/images/bus.jpg'
image_bytes = requests.get(image_url).content
image_path = os.path.join(output_dir, 'sample_image.jpg')
with open(image_path, 'wb') as f:
  f.write(image_bytes)
# Step 3: Inference / Object Detection run karein
results = model(image_path)
# Step 4: Bounding boxes aur labels process karein
for result in results:
  annotated_frame = result.plot()
  print("\n--- Detection Results ---")
  for box in result.boxes:
    cls_id = int(box.cls[0])
    label = result.names[cls_id]
    confidence = float(box.conf[0])
    bbox = box.xyxy[0].tolist()  # [xmin, ymin, xmax, ymax]
    print(f"Object: {label} | Confidence: {confidence:.2f} | BBox: {[round(x, 1) for x in bbox]}")
# Step 5: Final output image ko experiment9 folder ke andar save karein
```

```
output_path = os.path.join(output_dir, 'output_detection.png')

output_rgb = cv2.cvtColor(annotated_frame, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(10, 6))

plt.imshow(output_rgb)

plt.axis('off')

plt.title('YOLOv8 Object Detection Result')

plt.savefig(output_path, bbox_inches='tight')

plt.close()

print(f"\nOutput saved successfully inside: {output_path}}")
```





Questions & Answers

Q1. Differentiate between image classification, object detection, and image segmentation.

- **Image Classification:** Predicts a single class label for the entire image without specifying locations (e.g., classifying an image as "dog").
- **Object Detection:** Identifies multiple objects in an image, assigns class labels, and determines their spatial locations using bounding boxes.
- **Image Segmentation:** Classifies every individual pixel in an image to delineate precise boundaries of objects (e.g., instance or semantic segmentation).

Q2. Explain the working principle of the YOLO (You Only Look Once) object detection algorithm.

YOLO formulates object detection as a single regression problem. It divides the input image into an $S \times S$grid. Each grid cell predicts bounding box coordinates, confidence scores, and class probabilities simultaneously in a single forward pass through a convolutional neural network.

Q3. Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and practical applications.

Feature	YOLO	SSD	Faster R-CNN
Speed	Very High (Real-time)	High	Moderate / Low
Accuracy	High	Moderate to High	Very High
Architecture	Single-stage grid regression	Single-stage multi-scale features	Two-stage (RPN + Classifier)
Applications	Real-time surveillance, autonomous driving	Mobile and embedded devices	High-precision medical imaging, offline analysis

---

Q4. What is a bounding box? Why is it important in object detection?

A bounding box is a rectangular enclosure around an object defined by coordinates such as $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$or $(x_{\text{center}}, y_{\text{center}}, \text{width}, \text{height})$. It is essential because it provides spatial localization, indicating where objects are situated within the image.

Q5. Explain the significance of confidence score and Intersection over Union (IoU) in object detection.

- **Confidence Score:** Reflects the model's certainty that a predicted box contains an object and that the predicted class label is correct.
- **Intersection over Union (IoU):** Evaluates the spatial overlap between the predicted bounding box (A) and the ground truth bounding box (B): $\text{IoU} = \frac{\text{Area of Overlap } (A \cap B)}{\text{Area of Union } (A \cup B)}$

Q6. What are the advantages of using pre-trained deep learning models over traditional image processing techniques?

- **Automated Feature Extraction:** Automatically learns hierarchical features rather than relying on manually designed descriptors like SIFT or HOG.
- **Robustness:** Handles variations in illumination, scale, pose, and occlusion much better.
- **Transfer Learning:** Allows models trained on large-scale datasets to be fine-tuned efficiently on specific downstream tasks.

Q7. Why are datasets such as MS COCO and Pascal VOC widely used for object detection research?

They provide standardized, diverse, and well-annotated benchmarks with complex real-world scenes across dozens of categories, enabling reproducible performance evaluation using metrics like mean Average Precision (mAP).

Q8. Mention five real-world applications of object detection in computer vision.

1. Autonomous Vehicles (obstacle and pedestrian detection)
2. Automated Video Surveillance and Security
3. Healthcare (lesion/tumor detection in radiological scans)
4. Smart Retail (cashierless checkout systems)
5. Industrial Quality Inspection (assembly line defect detection)

Q9. What challenges are commonly encountered while detecting objects in complex real-world environments?

- Severe object occlusions and dense crowding
- Drastic variations in lighting and weather conditions
- Small or low-resolution objects with limited features
- High intra-class variance and aspect ratio shifts

Q10. How can object detection systems be further improved using recent advancements in deep learning and computer vision?

- Integrating vision-language transformers (e.g., Grounding DINO) for open-vocabulary detection.
- Model compression techniques (Quantization, Pruning, TensorRT) for fast edge device deployment.
- Leveraging temporal contextual features in video streaming for better tracking consistency.